

PROJECT PLANNING AND SCHEDULING REPORT

Construction of a Residential House

Plot 97, Block-D, Gulberg Greens, Islamabad

Prepared by

Hasnain Haroon

BS Civil Engineering

Department of Civil Engineering, COMSATS University Islamabad, Wah Campus

Course: Project Planning and Management (MGT462)

June 2025

Abstract

This report presents a complete planning and scheduling exercise for the construction of a 10-Marla, double-storey residential house located at Plot 97, Block-D, Gulberg Greens, Islamabad. The schedule was developed in Primavera P6 under a labour-only contract framework, with a fixed project duration of twelve months. Quantity estimates were derived from the architectural drawings, and productivity benchmarks were referenced from established industry sources, including PlanningPlanet.com. The project was organised through a detailed Enterprise Project Structure (EPS), Organisational Breakdown Structure (OBS), Work Breakdown Structure (WBS), and Resource Breakdown Structure (RBS), aligning every construction activity with a responsible project role. Scheduling logic incorporated Finish-to-Start (FS) and Start-to-Start (SS) relationships to enable concurrent execution of interior and exterior works. Primavera P6 was used to integrate scheduling constraints, resource histograms, resource levelling, and performance-tracking tools such as Earned Value Analysis (EVA) and S-curves, combining academic rigour with a realistic, practice-oriented project environment.

TABLE OF CONTENTS

1. Introduction	1
1.1 Overview	1
1.2 Statement of Purpose	1
1.3 Significance	2
1.4 Objectives	2
1.5 Limitations	2
2. Research Procedure	3
3. Project Planning and Scheduling Components	4
3.1 Enterprise Project Structure (EPS)	4
3.2 Organisational Breakdown Structure (OBS)	4
3.3 Work Breakdown Structure (WBS)	4
3.4 Resource Breakdown Structure (RBS)	5
3.5 Gantt Chart / Scheduling Model	5
3.6 Activity List and Duration Estimates	6
3.7 Earned Value Analysis (EVA)	6
3.8 Activity Usage Profile	7
3.9 Resource Usage Profile	7
3.10 Identified Risks	8
4. Conclusion	9
5. Recommendations	9
6. References	10

List of Figures

Figure 1: Enterprise Project Structure (EPS)	4
Figure 2: Work Breakdown Structure (WBS)	5
Figure 3: Gantt Chart / Scheduling Report	6
Figure 4: Activity List with Logical Relationships	6
Figure 5: Resource Usage Profile	8
Figure 6: Risk Register	8

1. Introduction

1.1 Overview

This report outlines the complete scheduling and planning process for the construction of a 10-Marla, double-storey residential house located in Islamabad. The project scope includes a fully functional home with six bedrooms, six bathrooms, two kitchens, living areas, a servant quarter, a storeroom, and a car porch. The schedule was developed using Primavera P6 to ensure timely and efficient project delivery within a fixed duration of twelve months.

The report presents a logical Work Breakdown Structure (WBS) that divides the construction into distinct phases, including site preparation, foundation works, superstructure, MEP installations, finishing works, external development, and final utility connections. Each phase is broken down into detailed activities with estimated durations, dependencies, and resource considerations. A Gantt chart and critical path analysis are included to visualise the sequence of activities and identify time-sensitive tasks.

Resource planning estimates labour requirements based on industry practices, while real-world constraints such as weather delays, supply chain issues, and regulatory factors are considered. Potential risks are identified along with corresponding mitigation strategies. The report concludes with reflections on the scheduling process, highlighting the importance of integrated planning, realistic estimation, and effective coordination for successful project execution.

1.2 Statement of Purpose

The purpose of this report is to plan and schedule the construction of a 10-Marla, double-storey residential house using Primavera P6. The objective is to create a realistic timeline that ensures timely and efficient project completion within a 12-month period. The project is broken down into logical phases through a Work Breakdown Structure (WBS), activity durations are estimated, and the critical path is identified using the Critical Path Method (CPM). Resource planning for labour and equipment is included while accounting for real-world constraints such as weather, regulatory issues, and supply-chain delays. By simulating a professional construction schedule, this report demonstrates the practical application of engineering planning and scheduling tools.

1.3 Significance

The significance of this report lies in its practical application of project scheduling principles to a real residential construction scenario. By using Primavera P6, the report demonstrates how structured planning, logical sequencing, and critical path analysis contribute to timely project delivery. It highlights the importance of integrating time, resources, and risk management to avoid delays and cost overruns, and helps in understanding the interdependency of construction activities and how effective coordination among teams ensures a smooth workflow. This exercise bridges the gap between theoretical project management concepts and their application in the construction industry.

1.4 Objectives

- Develop a detailed and logical Work Breakdown Structure (WBS) that clearly defines all phases and activities of the construction project.
- Estimate realistic activity durations using standard productivity rates and construction benchmarks.
- Assign labour roles and responsibilities within a labour-only resource allocation framework to simulate workforce planning.
- Integrate project-specific constraints such as weather delays, regulatory approvals, and working-hour calendars.
- Apply concurrent engineering principles to allow overlapping of tasks and improve schedule efficiency.
- Generate resource histograms to visualise labour demand and apply resource levelling techniques to avoid overallocation.
- Establish a framework for earned value management, enabling performance tracking through SPI, CPI, and S-curve analysis.

1.5 Limitations

- The total project duration was constrained to a maximum of 12 months as per the assumed client requirement.
- Material quantities were approximated based on standard residential construction norms in the absence of a formal Bill of Quantities (BOQ).
- Productivity rates used for estimating activity durations were referenced from credible industry sources, including PlanningPlanet.com.
- The project followed a labour-only contract model, which limited resource planning to manpower without accounting for material or equipment logistics.

2. Research Procedure

The methodology adopted for this scheduling exercise followed a structured sequence of analysis, planning, and modelling within Primavera P6:

- The architectural plans were thoroughly analysed to understand space utilisation and structural detailing.
- A structured Work Breakdown Structure (WBS) was developed to organise project activities into logical phases.
- Activity durations were calculated using recognised productivity benchmarks from construction databases.
- The WBS and activity sequences were input into Primavera P6, where interdependencies and logical links were defined.
- Labour resources were assigned to activities, aligning manpower with task requirements under the labour-only contract model.
- Appropriate relationships, including Start-to-Start (SS) and Finish-to-Start (FS), were used to enhance scheduling efficiency.
- Project-specific calendars and constraints were applied to control working hours and ensure proper sequencing.
- Resource histograms were generated and levelling techniques applied to balance labour allocation throughout the timeline.
- Earned Value Management (EVM) tools, including S-curves, were established to monitor progress and cost performance over the project lifecycle.

3. Project Planning and Scheduling Components

3.1 Enterprise Project Structure (EPS)

The Enterprise Project Structure (EPS) is a strategic framework used in Primavera P6 to manage multiple projects within an organisation. In this exercise, the residential construction project was categorised under the “Private Residential Projects” sector within the scheduling database, sitting under Engineering & Construction within an enterprise-wide hierarchy that also includes Commercial, Infrastructure, and Internal Projects. This structure enables centralised tracking, standardises scheduling practices, and supports better reporting and control across multiple ongoing developments.

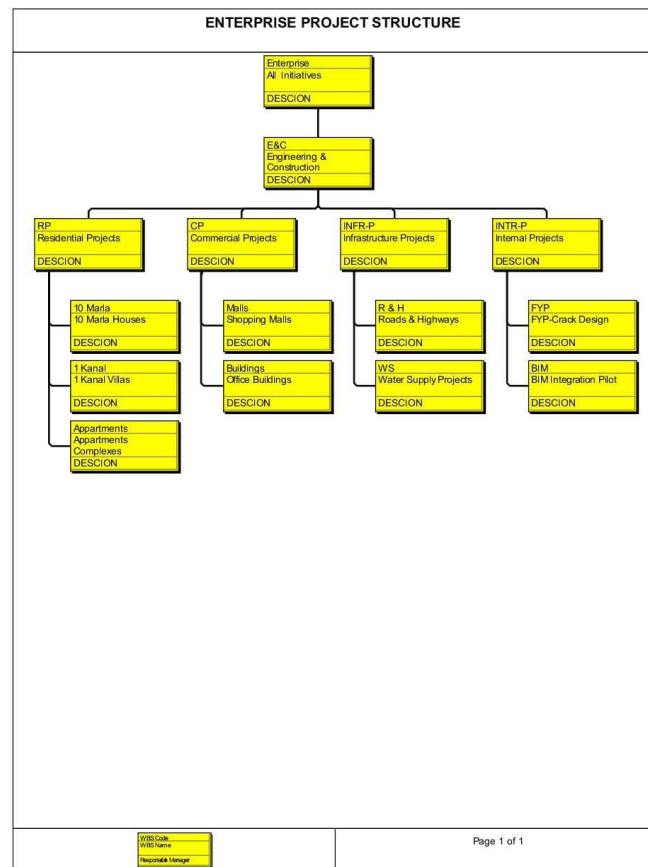


Figure 1: Enterprise Project Structure (EPS)

3.2 Organisational Breakdown Structure (OBS)

The OBS serves as the project's organisational map, outlining which team members are responsible for each segment of the work. Key roles assigned for this project include a Project Coordinator, Site Supervisor, and Quality Control Officer, each linked to specific activities such as layout execution, formwork inspection, and finishing-quality checks. This structure ensures proper oversight and accountability, helping to maintain safety, quality, and schedule compliance throughout the construction process.

3.3 Work Breakdown Structure (WBS)

The WBS divides the construction project into clearly defined, manageable phases. Major sections include site preparation, foundation works, superstructure (ground and first floor), MEP works, finishing works, external development, and project closure. Each WBS level was directly mapped to specific activities in Primavera P6, enabling logical sequencing and close alignment with the architectural drawings and the overall construction workflow.

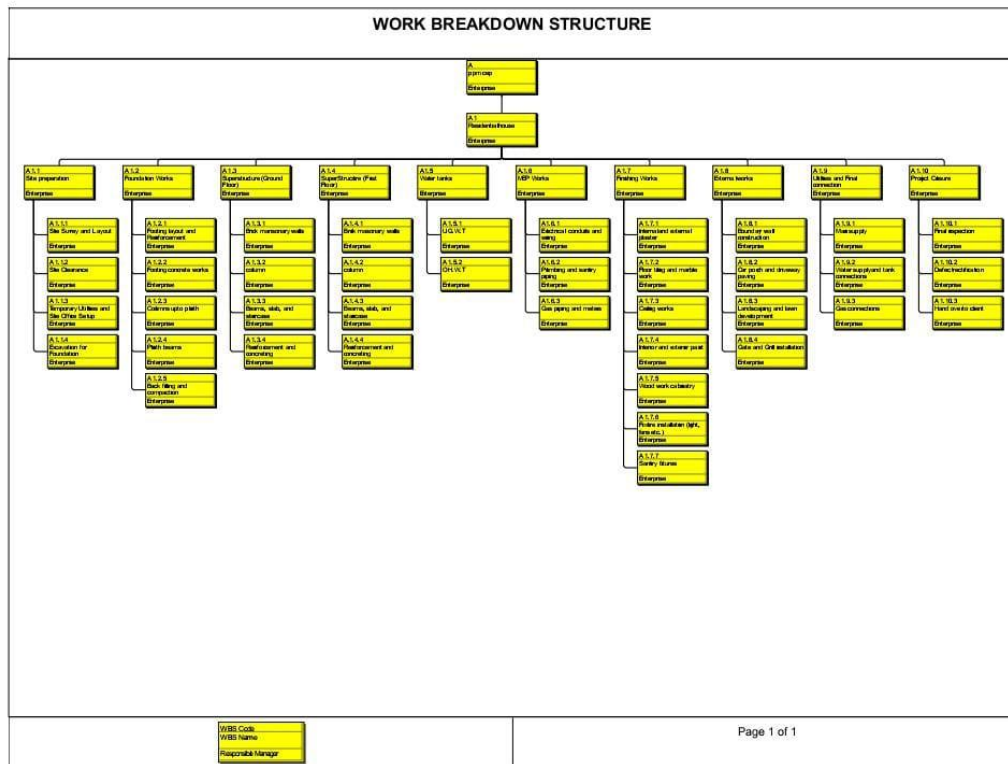


Figure 2: Work Breakdown Structure (WBS)

3.4 Resource Breakdown Structure (RBS)

The Resource Breakdown Structure organises labour and tools according to their relevance to project activities. Skilled and unskilled labour categories were created, including masons, carpenters, steel fixers, painters, and electricians, paired with essential equipment such as mixers, vibrators, and scaffolding. Resources were allocated according to task requirements — for example, plastering crews assigned to wall finishes — and managed to optimise efficiency and avoid bottlenecks during peak workload periods.

3.5 Gantt Chart / Scheduling Model

A detailed Gantt chart was generated using Primavera P6 to represent the full project timeline, spanning from January 2025 to January 2026. The chart breaks the schedule into logical construction stages and illustrates start and finish dates, dependencies, and critical path activities. Critical early tasks, such as foundation pouring and frame erection, were identified as schedule drivers, with overlapping interior and

exterior works planned to improve overall time efficiency. This model helped visualise progress and allocate resources in a coordinated manner.

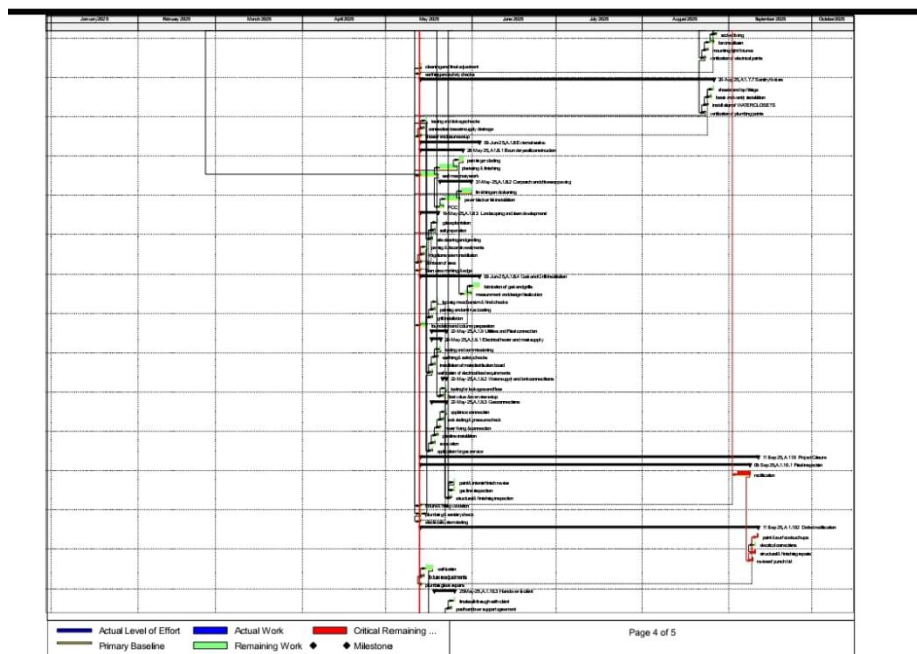
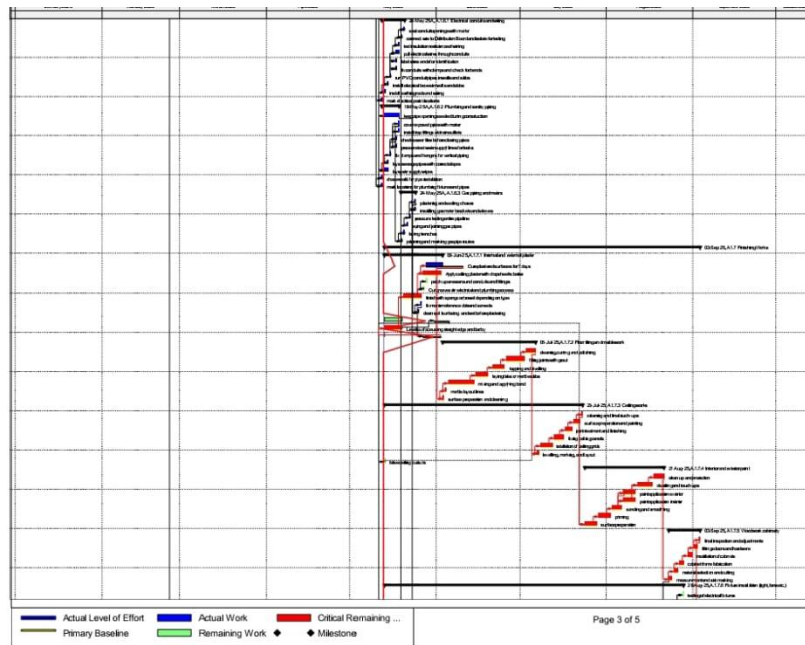


Figure 3: Gantt Chart / Scheduling Report

3.6 Activity List and Duration Estimates

An extensive list of over 300 construction activities was created, each with logical predecessors and successors. Durations were estimated using standard productivity benchmarks and cross-verified against real-world practices. The sequence ensures fluid transitions from task to task — for example, from slab curing to blockwork, and then to plastering. Each activity's duration reflects both technical complexity and labour capacity, making the overall schedule practical and achievable.

Activity ID	Activity Name	Duration (Days)	Start Date	End Date	Predecessor(s)	Successor(s)	Resource(s)	Material(s)	Cost (PKR)	Notes	Remarks	Actual Start	Actual End	Actual Duration (Days)	Actual Cost (PKR)	Actual Remarks	Actual Duration (Days)	Actual Cost (PKR)	Actual Remarks
A.1.1.1	Site clearing and leveling	1	2023-07-01	2023-07-01					1000										
A.1.1.2	Foundation excavation	4	2023-07-02	2023-07-05					4000										
A.1.1.3	Foundation concrete pouring	1	2023-07-06	2023-07-06					1000										
A.1.1.4	Foundation curing	7	2023-07-07	2023-07-13					7000										
A.1.1.5	Foundation reinforcement	1	2023-07-14	2023-07-14					1000										
A.1.1.6	Foundation backfilling	1	2023-07-15	2023-07-15					1000										
A.1.1.7	Foundation inspection	1	2023-07-16	2023-07-16					1000										
A.1.1.8	Foundation completion	1	2023-07-17	2023-07-17					1000										
A.1.1.9	Foundation curing	7	2023-07-18	2023-07-24					7000										
A.1.1.10	Foundation reinforcement	1	2023-07-25	2023-07-25					1000										
A.1.1.11	Foundation backfilling	1	2023-07-26	2023-07-26					1000										
A.1.1.12	Foundation inspection	1	2023-07-27	2023-07-27					1000										
A.1.1.13	Foundation completion	1	2023-07-28	2023-07-28					1000										
A.1.1.14	Foundation curing	7	2023-07-29	2023-08-04					7000										
A.1.1.15	Foundation reinforcement	1	2023-08-05	2023-08-05					1000										
A.1.1.16	Foundation backfilling	1	2023-08-06	2023-08-06					1000										
A.1.1.17	Foundation inspection	1	2023-08-07	2023-08-07					1000										
A.1.1.18	Foundation completion	1	2023-08-08	2023-08-08					1000										
A.1.1.19	Foundation curing	7	2023-08-09	2023-08-15					7000										
A.1.1.20	Foundation reinforcement	1	2023-08-16	2023-08-16					1000										
A.1.1.21	Foundation backfilling	1	2023-08-17	2023-08-17					1000										
A.1.1.22	Foundation inspection	1	2023-08-18	2023-08-18					1000										
A.1.1.23	Foundation completion	1	2023-08-19	2023-08-19					1000										
A.1.1.24	Foundation curing	7	2023-08-20	2023-08-26					7000										
A.1.1.25	Foundation reinforcement	1	2023-08-27	2023-08-27					1000										
A.1.1.26	Foundation backfilling	1	2023-08-28	2023-08-28					1000										
A.1.1.27	Foundation inspection	1	2023-08-29	2023-08-29					1000										
A.1.1.28	Foundation completion	1	2023-08-30	2023-08-30					1000										
A.1.1.29	Foundation curing	7	2023-08-31	2023-09-06					7000										
A.1.1.30	Foundation reinforcement	1	2023-09-07	2023-09-07					1000										
A.1.1.31	Foundation backfilling	1	2023-09-08	2023-09-08					1000										
A.1.1.32	Foundation inspection	1	2023-09-09	2023-09-09					1000										
A.1.1.33	Foundation completion	1	2023-09-10	2023-09-10					1000										
A.1.1.34	Foundation curing	7	2023-09-11	2023-09-17					7000										
A.1.1.35	Foundation reinforcement	1	2023-09-18	2023-09-18					1000										
A.1.1.36	Foundation backfilling	1	2023-09-19	2023-09-19					1000										
A.1.1.37	Foundation inspection	1	2023-09-20	2023-09-20					1000										
A.1.1.38	Foundation completion	1	2023-09-21	2023-09-21					1000										
A.1.1.39	Foundation curing	7	2023-09-22	2023-09-28					7000										
A.1.1.40	Foundation reinforcement	1	2023-09-29	2023-09-29					1000										
A.1.1.41	Foundation backfilling	1	2023-09-30	2023-09-30					1000										
A.1.1.42	Foundation inspection	1	2023-10-01	2023-10-01					1000										
A.1.1.43	Foundation completion	1	2023-10-02	2023-10-02					1000										
A.1.1.44	Foundation curing	7	2023-10-03	2023-10-09					7000										
A.1.1.45	Foundation reinforcement	1	2023-10-10	2023-10-10					1000										
A.1.1.46	Foundation backfilling	1	2023-10-11	2023-10-11					1000										
A.1.1.47	Foundation inspection	1	2023-10-12	2023-10-12					1000										
A.1.1.48	Foundation completion	1	2023-10-13	2023-10-13					1000										
A.1.1.49	Foundation curing	7	2023-10-14	2023-10-20					7000										
A.1.1.50	Foundation reinforcement	1	2023-10-21	2023-10-21					1000										
A.1.1.51	Foundation backfilling	1	2023-10-22	2023-10-22					1000										
A.1.1.52	Foundation inspection	1	2023-10-23	2023-10-23					1000										
A.1.1.53	Foundation completion	1	2023-10-24	2023-10-24					1000										
A.1.1.54	Foundation curing	7	2023-10-25	2023-10-31					7000										
A.1.1.55	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.56	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.57	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.58	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.59	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.60	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.61	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.62	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.63	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.64	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.65	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.66	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.67	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.68	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.69	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.70	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.71	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.72	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.73	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.74	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.75	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.76	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.77	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.78	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.79	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.80	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.81	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.82	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.83	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.84	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.85	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.86	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.87	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.88	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.89	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.90	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										
A.1.1.91	Foundation backfilling	1	2023-10-33	2023-10-33					1000										
A.1.1.92	Foundation inspection	1	2023-10-34	2023-10-34					1000										
A.1.1.93	Foundation completion	1	2023-10-35	2023-10-35					1000										
A.1.1.94	Foundation curing	7	2023-10-36	2023-10-31					7000										
A.1.1.95	Foundation reinforcement	1	2023-10-32	2023-10-32					1000										

3.7 Earned Value Analysis (EVA)

Earned Value Analysis was conducted to evaluate project performance in terms of cost and time. As of the latest monitoring date, the project showed a slightly negative variance, with the Earned Value trailing behind the Planned Value and Actual Cost. A Schedule Performance Index (SPI) of 0.94 indicated that the project was slightly behind schedule, while a Cost Performance Index (CPI) of 0.99 reflected effective cost control. Delays in finishing tasks, such as electrical fittings and interior painting, were identified as contributing factors, underscoring the importance of timely coordination and labour availability under a fixed project deadline.

3.8 Activity Usage Profile

The Activity Usage Profile illustrates how project costs are distributed over time, providing a visual representation of planned, earned, and actual expenditure. The profile typically displays multiple curves — Budgeted Total Cost, Actual Total Cost, Planned Value, Earned Value, and Remaining Cost. In this project, the profile closely resembles a classic S-curve, a common pattern in construction that starts slowly, intensifies during the structural and finishing phases, and slows near completion. This progression reflects increased activity and complexity during key stages such as superstructure works and final finishes, as outlined in the WBS.

The S-curve plays a crucial role in evaluating project performance. A close alignment between the Planned Value and Earned Value curves indicates smooth progress, while significant deviations suggest delays or inefficiencies. A flatter or delayed curve can point to late starts or rework in phases such as MEP installations or exterior finishes. Given the strict 12-month deadline, such delays place pressure on later tasks, making schedule compression and resource coordination increasingly critical as the project advances.

3.9 Resource Usage Profile

The Resource Usage Profile displays how labour and equipment are utilised across the entire project timeline. The chart includes Budgeted Units, Actual Units, Overallocated Units, and defined Resource Limits. Forming a bell-shaped curve, this profile indicates a gradual resource build-up, a peak during intensive construction activities, and a decline as the project nears completion — aligning with peak workload phases such as structure completion, MEP installations, and interior finishes.

This profile is instrumental in monitoring workforce distribution, ensuring that trades such as masons, electricians, steel fixers, and painters are effectively deployed without overloading any single team. The absence of sharp peaks in overallocation areas suggests efficient scheduling and coordination of concurrent trades; however, any noticeable spike in resource overuse — especially during high-activity months — may indicate issues such as rework, delays in material delivery, or the need to recover lost time from earlier setbacks. The overall sequencing, from site preparation through to handover, matches Primavera's logic and aligns with realistic architectural milestones such as slab casting, utility connections, and final inspection.

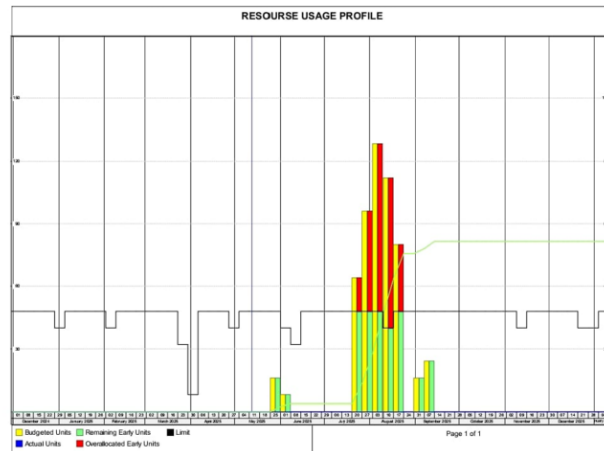


Figure 5: Resource Usage Profile

3.10 Identified Risks

A formal risk register was maintained within Primavera P6 to track potential threats to the project schedule, cost, and quality. Each risk was assigned an owner, an identification date, and an exposure window, allowing the project team to monitor and respond to issues proactively throughout the construction period.

Risk ID	Risk Name	Category	Status	Owner
RISK1	Delay in project approval	Planning	Proposed	PM / General Manager
RISK2	Site or material delivery delay	Performance & Risk	Active	Site Engineer
RISK3	Unexpected soil conditions	Estimating	Proposed	QC Engineer

Risk ID	Risk Name	Category	Status	Owner
RISK4	Labour shortage	Resources	Active	Site Engineer
RISK5	Design scope changes	Complexity & Intent	Managed (Closed)	Site Engineer
RISK6	Unfavourable weather	Weather	Proposed	PM / Site Engineer

Figure 6: Risk Register

4. Conclusion

This report demonstrates the systematic planning and scheduling of a 10-Marla, double-storey residential house using Primavera P6. Through the development of a structured Work Breakdown Structure (WBS), logical activity sequencing, and realistic duration estimates, a detailed and optimised schedule was created to meet a 12-month deadline. The integration of resource planning, constraint management, and earned-value tracking provided valuable insight into project performance and control.

Visual tools such as Gantt charts, S-curves, and resource usage profiles supported effective decision-making and highlighted critical areas requiring attention. The use of Primavera P6 allowed for efficient modelling of real-world construction complexities, reinforcing the importance of coordination, early planning, and risk mitigation in residential project management. Overall, the exercise strengthened both technical and analytical planning skills, aligning theoretical knowledge with professional construction practice.

5. Recommendations

- Develop a formal Bill of Materials (BOM) for future iterations to replace approximated material quantities with verified cost and quantity data.
- Build buffer time into finishing-stage activities (electrical fittings, interior painting) to reduce the schedule slippage reflected in the SPI of 0.94.
- Apply resource levelling more aggressively during peak months (July–September) to flatten the overallocation spikes observed in the resource usage profile.
- Introduce a formal change-control process to manage design scope changes before they affect the critical path.
- Schedule weather-sensitive activities, such as concrete pouring and external finishing, outside the monsoon season where possible to reduce weather-related risk exposure.
- Conduct periodic Earned Value Analysis updates (monthly) to track SPI and CPI trends and trigger corrective action earlier in the project lifecycle.

6. References

PlanningPlanet. (n.d.). Construction productivity benchmarks and planning resources. Retrieved from <https://www.planningplanet.com>

Oracle Corporation. (2024). Primavera P6 Professional Project Management — User's Guide. Oracle Corporation.

Project Management Institute. (2021). A Guide to the Project Management Body of Knowledge (PMBOK Guide) (7th ed.). Project Management Institute.